

test-coverage

LLMProvider Test Coverage Ledger (round-292, vasic- digital twin)

This ledger maps the surface-area symbols of LLMProvider to the specific tests + Challenges that exercise them with positive runtime evidence (CONST-035 / Article XI §11.9, CONST-050(B)).

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17): "all existing tests and Challenges do work in anti-bluff manner

- they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

Anti-bluff guarantees

Every row below identifies BOTH a symbol AND the failing-line a mutation would produce. Mutation pairing per §1.1 / CONST-050(A).

Ledger — pkg/circuit

Symbol	Test	Challenge invariant
NewCircuitBreaker()	pkg/circuit/ circuit_breaker_test.go	runner construction p
NewDefaultCircuitBreaker()	pkg/circuit/ circuit_breaker_test.go	(covered via NewCirc
DefaultCircuitBreakerConfig()		

Symbol	Test	Challenge invariant
	pkg/circuit/ circuit_breaker_test.go	runner uses defaults of fixture
CircuitBreaker.GetState()	pkg/circuit/ circuit_breaker_test.go	circuit.initial_state.
CircuitBreaker.IsClosed()/ IsOpen()	pkg/circuit/ circuit_breaker_test.go	circuit.is_closed.<loc
CircuitBreaker.Complete()	pkg/circuit/ circuit_breaker_test.go	circuit.opens_after_fa
CircuitBreaker.ErrCircuitOpen	pkg/circuit/ circuit_breaker_test.go	runner asserts errors. ErrCircuitOpen)
CircuitBreaker.AddListener()	pkg/circuit/ circuit_breaker_test.go	(covered by unit suite)
CircuitBreakerManager.Register()	pkg/circuit/ circuit_breaker_test.go	(covered by unit suite)

Ledger — pkg/health

Symbol	Test	Challenge invariant
NewHealthMonitor() / Default*	pkg/health/ health_monitor_test.go	runner constructs one monitor
HealthMonitor.RegisterProvider()	pkg/health/ health_monitor_test.go	health.initial_status.< (x5)
HealthMonitor.GetHealth()	pkg/health/ health_monitor_test.go	runner asserts Status==Unknown init
HealthMonitor.RecordFailure()	pkg/health/ health_monitor_test.go	health.transitions_afte
HealthMonitor.AddListener()	pkg/health/ health_monitor_test.go	runner counts listener
HealthStatusUnknown/Unhealthy constants	pkg/health/ health_monitor_test.go	runner asserts status s als

Ledger — pkg/retry

Symbol	Test	Challenge invariant
IsRetryableStatusCode()	pkg/retry/ retry_test.go	retry.is_retryable_status_code
CalculateBackoff()	pkg/retry/ retry_test.go	retry.calculate_backoff_bounds
DefaultRetryConfig()	pkg/retry/ retry_test.go	runner builds custom config to bound MaxDelay
ExecuteWithRetry()		

Symbol	Test	Challenge invariant
	pkg/retry/ retry_test.go	(covered by unit + HTTP suites)
RetryConfig struct fields	pkg/retry/ retry_test.go	runner validates Initial/Max/ Multiplier bounds

Ledger — pkg/provider + pkg/models

Symbol	Test	Challenge invariant
provider.LLMProvider iface	(every provider package)	runner satisfies it via controllableProvider
models.LLMRequest	pkg/models/ types_test.go	runner builds real requests per-fixture
models.LLMResponse	pkg/models/ types_test.go	runner threads through breaker
models.ProviderCapabilities	pkg/models/ types_test.go	runner asserts non- nil from controllable

Challenges (challenges/ + challenges/ scripts/)

Challenge	Purpose	Mutation-paired?
llmprovider_describe_challenge.sh	runner in- variants (23 PASS)	YES (round-292)
chaos_failure_injection_challenge.sh	runtime chaos resili- ence	env-gated
ddos_health_flood_challenge.sh	DDoS health-flood resilience	env-gated
scaling_horizontal_challenge.sh	horizontal scale check	env-gated
stress_sustained_load_challenge.sh	stress floor	env-gated
ui_terminal_interaction_challenge.sh	TUI smoke	env-gated
ux_end_to_end_flow_challenge.sh	UX end-to- end	env-gated
host_no_auto_suspend_challenge.sh		n

Challenge	Purpose	Mutation-paired?
	CONST-033 host-power ban	
no_suspend_calls_challenge.sh	CONST-033 anti-sus- pend scan	n

How to run

```
# Race-detector full unit + runner pass:
GOMAXPROCS=2 go test -count=1 -race -p 1 ./...

# Anti-bluff describe Challenge (23 invariants):
./challenges/llmprovider_describe_challenge.sh normal # exit 0
./challenges/llmprovider_describe_challenge.sh mutate # exit 99

# Standalone runner (useful for forensics):
go run ./challenges/runner/
LLMPROVIDER_MUTATE_RUNNER=1 go run ./challenges/runner/
```

Cascade

CONST-047 + CONST-051(A): this ledger is the LLMProvider-side realisation of the cross-org "equal codebase" mandate. The vasic-digital twin's round-292 enrichment mirrors the HelixDevelopment twin's round-276 work; the two twins share source-code parity in pkg/circuit / pkg/health / pkg/retry / pkg/models but maintain divergent git histories per the round-276 finding. Sibling sub-modules carry the same ledger pattern: LeakHub (round 266), MCP_Module (round 267), Models (round 268), Ouroborous (round 269), Planning (round 270), conversation (round 271), DebateOrchestrator (round 272), HelixSpecifier (round 273), TOON (round 286), VectorDB (round 287), Veritas (round 288), Watcher (round 289). Bumped from the meta-repo on every round close-out.